

TASK 3– DATA ANALYSIS USING PANDAS

1. Student Details

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- Internship Domain: Python developer
- Date:23-05-2026

2. Task Objective

The objective of this task is to perform data analysis using Python and Pandas. The task includes loading a CSV dataset, cleaning missing data, applying filtering and grouping operations, and extracting meaningful insights.

3. Tools and Technologies Used

- Python
- Pandas
- Visual Studio Code

4. Dataset Information

- Dataset Name: data
- File Format: CSV
- Number of Rows:1201
- Number of Columns:06
- Dataset Description: It is about the mental health of Teens

```
1 | #gender,daily_social_media_hours,platform_usage,sleep_hours,screen_time_before_sleep,academic_performance,physical_activity,social_interaction_level,stress_level,society_level,addiction_level,depression_label
2 | 14,male,7.9,Instagram,7.4,2.5,0.0,1.5,low,2.2,1.0
3 | 15,female,1.9,TikTok,8.0,2.9,3.22,0.0,high,0.1,10.0
4 | 17,female,1.1,Instagram,7.6,0.5,3.92,0.0,high,2.4,2.0
5 | 15,male,7.4,TikTok,6.9,1.6,3.40,0.0,medium,1.2,0.0
6 | 15,female,4.7,Both,4.0,1.0,2.37,1.4,medium,3.5,2.0
7 | 19,female,7.4,Both,4.4,2.4,2.63,0.0,high,3.5,7.0
8 | 16,female,2.3,Instagram,6.4,2.4,2.0,0.7,low,2.2,3.0
9 | 16,male,4.0,Both,4.2,0.5,2.4,1.1,low,6.10,5.0
10 | 19,female,3.1,TikTok,5.0,2.1,2.00,0.9,high,1.10,5.0
11 | 15,male,1.9,TikTok,6.9,2.5,3.77,1.1,high,1.1,4.0
12 | 16,male,7.2,Both,5.8,2.5,3.16,1.2,medium,5.0,5.0
13 | 11,male,4.9,Instagram,4.7,2.5,1.92,1.7,low,1.10,0.0
14 | 16,male,6.2,Both,7.2,2.2,0.2,1.7,medium,7.7,0.0
15 | 17,female,6.0,TikTok,5.7,1.1,3.79,1.3,medium,1.10,10.0
16 | 15,male,5.7,TikTok,5.9,0.0,3.5,0.7,low,3.9,1.0
17 | 14,female,3.4,Both,5.3,0.6,2.99,0.3,high,6.10,2.0
18 | 11,male,1.5,Both,4.2,1.4,2.59,0.5,high,4.5,0.0
19 | 16,male,1.1,Both,6.1,0.0,2.11,1.9,high,10.7,10.0
20 | 16,female,6.2,Both,6.0,1.9,3.00,1.1,high,10.9,1.0
21 | 14,female,1.2,Instagram,7.4,1.0,3.06,0.0,high,8.4,6.0
22 | 16,male,1.4,TikTok,4.6,2.1,2.24,0.4,high,4.1,1.0
23 | 16,male,4.1,Both,7.4,2.2,0.2,0.1,medium,6.4,0.0
24 | 18,male,5.3,Instagram,4.9,1.2,3.09,1.2,low,5.4,5.0
25 | 18,female,2.1,TikTok,7.0,0.5,2.96,1.7,medium,9.0,7.0
26 | 16,female,2.4,Both,6.7,2.0,2.53,0.1,high,0.0,0.0
27 | 13,female,1.5,Both,7.2,2.5,2.15,1.3,medium,4.0,2.0
28 | 14,female,5.7,Both,8.7,0.6,2.64,0.1,low,8.9,4.0
29 | 19,male,2.0,TikTok,7.1,0.9,2.26,1.0,high,7.5,0.0
30 | 14,male,2.5,TikTok,5.6,2.9,2.24,0.5,low,3.9,5.0
31 | 16,female,3.4,TikTok,7.0,2.0,3.1,1.0,medium,6.6,4.0
32 | 15,male,6.4,TikTok,5.7,2.5,2.07,1.1,low,2.1,10.0
33 | 11,female,1.4,Instagram,8.7,2.5,3.09,0.4,high,8.2,9.0
34 | 16,male,2.7,Instagram,4.2,2.4,3.05,0.2,high,1.0,4.0
35 | 16,male,4.1,TikTok,8.0,0.0,2.11,0.0,low,6.4,7.0
36 | 13,female,6.0,TikTok,7.1,1.6,3.77,0.1,medium,10.2,1.0
37 | 17,female,5.0,Both,4.9,2.1,2.01,1.0,low,5.4,5.0
38 | 19,female,6.0,Instagram,5.2,2.5,3.96,0.0,high,3.10,7.0
39 | 17,female,4.0,Instagram,4.4,1.3,2.02,0.5,low,5.10,6.0
40 | 17,female,1.3,TikTok,8.0,2.0,2.8,1.0,high,1.7,0.0
41 | 12,female,6.3,Instagram,4.0,1.0,2.03,1.2,high,1.10,0.0
42 | 14,male,4.4,Instagram,5.7,0.5,2.00,1.0,high,10.4,2.0
43 | 19,male,1.4,Instagram,8.4,1.4,3.01,0.4,low,10.4,1.0
44 | 16,female,3.3,Instagram,7.2,1.4,3.40,0.3,medium,6.1,0.0
```

5. Code Implementation

Step 1: Import Libraries

```
import pandas as pd
```

Step 2: Load Dataset

```
# Load dataset
```

```
df = pd.read_csv("data.csv")
```

```
# Show first 5 rows
```

```
print("FIRST 5 ROWS")
```

```
print(df.head())
```

```
# Dataset info
```

```
print("\nDATASET INFO")
```

```
print(df.info())
```

```
FIRST 5 ROWS
  age  gender  daily_social_media_hours  platform_usage  ...  stress_level  anxiety_level  addiction_level  depression_label
0   14   male                7.9      Instagram  ...         2             2             1             0
1   19  female                1.9        TikTok  ...         8             1            10             0
2   17  female                1.3      Instagram  ...         2             4             2             0
3   15   male                7.4        TikTok  ...         1             7             9             0
4   15  female                4.7          Both  ...         3             5             2             0

[5 rows x 13 columns]
```

Step 3: Data Cleaning

```
# Missing values
```

```
print("\nMISSING VALUES")
```

```
print(df.isnull().sum())
```

```
# Remove missing values
```

```
df = df.dropna()
```

```
# Remove duplicates
```

```
df = df.drop_duplicates()
```

```
MISSING VALUES
age                0
gender             0
daily_social_media_hours  0
platform_usage     0
sleep_hours        0
screen_time_before_sleep  0
academic_performance  0
physical_activity   0
social_interaction_level  0
stress_level        0
anxiety_level       0
addiction_level     0
depression_label    0
dtype: int64
```

Step 4: Filtering and Grouping

Example filtering

```
print("filtering")
```

```
filtered = df[df["age"] < 20]
```

```
print(filtered)
```

```
filtering
   age  gender  daily_social_media_hours  platform_usage  ...  stress_level  anxiety_level  addiction_level  depression_label
0    14   male         7.9      Instagram  ...         2             2             1             0
1    19  female         1.9        TikTok  ...         8             1            10             0
2    17  female         1.3      Instagram  ...         2             4             2             0
3    15   male         7.4        TikTok  ...         1             7             9             0
4    15  female         4.7         Both  ...         3             5             2             0
...   ...   ...         ...         ...  ...         ...             ...             ...             ...
1195  18  female         6.8      Instagram  ...         3             4             4             0
1196  16   male         2.3         Both  ...         7             4             4             0
1197  14  female         1.7         Both  ...         1             1             1             0
1198  15   male         3.9         Both  ...         7             9             9             0
1199  16  female         4.7        TikTok  ...         5             7             3             0

[1200 rows x 13 columns]
```

Example grouping

```
print("grouping")
```

```
grouped = df.groupby("age")["sleep_hours"].mean()
```

```
print(grouped)
```

```
grouping
age
13    6.485500
14    6.315686
15    6.493889
16    6.485890
17    6.505882
18    6.359884
19    6.480864
Name: sleep_hours, dtype: float64

Data cleaned successfully!
PS C:\Users\GuruPavan\OneDrive\Desktop\pandas\internship_task>
```

Step 5: Statistical Analysis

```
# Statistics
```

```
print("\nSTATISTICS")
```

```
print(df.describe())
```

STATISTICS							
	age	daily_social_media_hours	sleep_hours	...	anxiety_level	addiction_level	depression_label
count	1200.000000	1200.000000	1200.000000	...	1200.000000	1200.000000	1200.000000
mean	15.928333	4.536667	6.449417	...	5.636667	5.565000	0.025833
std	2.021947	2.029599	1.442677	...	2.859453	2.830627	0.158704
min	13.000000	1.000000	4.000000	...	1.000000	1.000000	0.000000
25%	14.000000	2.800000	5.200000	...	3.000000	3.000000	0.000000
50%	16.000000	4.500000	6.500000	...	6.000000	6.000000	0.000000
75%	18.000000	6.300000	7.600000	...	8.000000	8.000000	0.000000
max	19.000000	8.000000	9.000000	...	10.000000	10.000000	1.000000

6. Insights / Findings

1. Missing values and duplicates were cleaned successfully.
2. Grouping analysis helped identify category-wise patterns.
3. Statistical analysis showed important trends in the dataset.
4. The dataset became more structured after preprocessing.
5. Data visualization made the analysis easier to understand.

7. Challenges Faced

- Handling missing values
- Understanding dataset columns
- Applying grouping and aggregation functions

8. Conclusion

This task helped in understanding practical data analysis using Pandas. It improved knowledge of data cleaning, filtering, grouping, aggregation, and visualization techniques using Python.

9. Project Links

GitHub Repository Link

https://github.com/tsgurupavankumar-creator/Alfido-python-developer/tree/95129fb9e771160d07e14421c66e37915f3f965a/pandas_internship_task

10. Output Screenshots

```
PS C:\Users\GuruPavan\OneDrive\Desktop\pandas_internship_task> python analysis.py
FIRST 5 ROWS
   age  gender  daily_social_media_hours  platform_usage  ...  stress_level  anxiety_level  addiction_level  depression_label
0   14   male              7.9      Instagram  ...          2              2              1              0
1   19  female              1.9        TikTok  ...          8              1             10              0
2   17  female              1.3      Instagram  ...          2              4              2              0
3   15   male              7.4        TikTok  ...          1              7              9              0
4   15  female              4.7          Both  ...          3              5              2              0

[5 rows x 13 columns]

DATASET INFO
<class 'pandas.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 13 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   age                                   1200 non-null   int64
 1   gender                               1200 non-null   str
 2   daily_social_media_hours             1200 non-null   float64
 3   platform_usage                       1200 non-null   str
 4   sleep_hours                         1200 non-null   float64
 5   screen_time_before_sleep            1200 non-null   float64
 6   academic_performance                1200 non-null   float64
 7   physical_activity                   1200 non-null   float64
 8   social_interaction_level             1200 non-null   str
 9   stress_level                        1200 non-null   int64
10   anxiety_level                       1200 non-null   int64
11   addiction_level                     1200 non-null   int64
12   depression_label                    1200 non-null   int64
dtypes: float64(5), int64(5), str(3)
memory usage: 122.0 KB
None

MISSING VALUES
age                0
gender             0
daily_social_media_hours  0
platform_usage     0
sleep_hours        0
screen_time_before_sleep  0
academic_performance  0
physical_activity   0
social_interaction_level  0
stress_level        0
anxiety_level       0
addiction_level     0
depression_label    0
dtype: int64

STATISTICS
   age  daily_social_media_hours  sleep_hours  ...  anxiety_level  addiction_level  depression_label
count  1200.000000              1200.000000  1200.000000  ...    1200.000000      1200.000000      1200.000000
mean    15.928333              4.536667      6.449417  ...      5.636667        5.565000        0.025833
std      2.021947              2.029599      1.442677  ...      2.859453        2.830627        0.158704
min     13.000000              1.000000      4.000000  ...      1.000000        1.000000        0.000000
25%     14.000000              2.800000      5.200000  ...      3.000000        3.000000        0.000000
50%     16.000000              4.500000      6.500000  ...      6.000000        6.000000        0.000000
75%     18.000000              6.300000      7.600000  ...      8.000000        8.000000        0.000000
max     19.000000              8.000000      9.000000  ...     10.000000       10.000000        1.000000
```

```

filtering
   age  gender  daily_social_media_hours  platform_usage  ...  stress_level  anxiety_level  addiction_level  depression_label
0    14   male          7.9          Instagram  ...          2             2             1             0
1    19  female          1.9             TikTok  ...          8             1            10             0
2    17  female          1.3          Instagram  ...          2             4             2             0
3    15   male          7.4             TikTok  ...          1             7             9             0
4    15  female          4.7             Both   ...          3             5             2             0
...   ...   ...          ...          ...   ...   ...   ...   ...   ...
1195  18  female          6.8          Instagram  ...          3             4             4             0
1196  16   male          2.3             Both   ...          7             4             4             0
1197  14  female          1.7             Both   ...          1             1             1             0
1198  15   male          3.9             Both   ...          7             9             9             0
1199  16  female          4.7             TikTok  ...          5             7             3             0

```

[1200 rows x 13 columns]

grouping

age

```

13    6.485500
14    6.315686
15    6.493889
16    6.485890
17    6.505882
18    6.359884
19    6.480864

```

Name: sleep_hours, dtype: float64

Data cleaned successfully!

PS C:\Users\GuruPavan\OneDrive\Desktop\pandas\internship_task>